DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

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ABSTRACT

We introduce our first-generation reasoning models, DeepSeek-R1-Zero and DeepSeek-R1. DeepSeek-R1-Zero, a model trained via large-scale reinforcement learning (RL) without supervised fine-tuning (SFT) as a preliminary step, demonstrates remarkable reasoning capabilities. Through RL, DeepSeek-R1-Zero naturally emerges with numerous powerful reasoning behaviors, including self-reflection, verification, and generating long chains of thought. However, R1-Zero encounters challenges such as poor readability and language mixing. To address these issues, we introduce DeepSeek-R1, which incorporates multi-stage training and cold-start data before RL. DeepSeek-R1 achieves performance comparable to OpenAI-o1-1217 on reasoning tasks, achieving 79.8% pass@1 on AIME 2024 and 97.3% on MATH-500. We also distill reasoning capabilities into smaller dense models (1.5B to 32B), which outperform comparable baselines.

1. Introduction & The Paradigm Shift in Reasoning

Recent advancements in large language models have pushed frontier capabilities, yet reasoning benchmarks like competitive mathematics and software synthesis remained challenging for standard next-token predictors. Rather than relying on massive supervised human demonstrations of reasoning, we explore pure reinforcement learning (RL) to incentivize self-discovery of reasoning paths.
DeepSeek-R1-Zero is trained directly on base models using Group Relative Policy Optimization (GRPO) without any prior supervised fine-tuning (SFT).

2. Group Relative Policy Optimization (GRPO) Mechanics

To eliminate the memory burden of a critic network in PPO, GRPO samples a group of outputs {o_1, o_2, ..., o_G} for each question q from the old policy pi_theta_old, and optimizes the policy using relative advantages computed solely within the group:
A_i = (r_i - mean(r)) / std(r).
The reward r_i combines rule-based accuracy rewards (e.g. verifying final mathematical answers in box format or code execution) and formatting rewards, completely removing the need for neural reward models.

3. The DeepSeek-R1 Multi-Stage Training Pipeline

To address language inconsistency and unstructured output in R1-Zero, DeepSeek-R1 adopts a four-stage pipeline:
Stage 1: Cold-start SFT with several thousand high-quality human-readable long chain-of-thought examples.
Stage 2: Reasoning-oriented RL using GRPO on math, logic, and coding benchmarks.
Stage 3: Rejection sampling to generate 800k synthetic reasoning and general-domain examples for comprehensive SFT.
Stage 4: Secondary RL with combined rule-based accuracy rewards and human preference alignment rewards.

3.1 Knowledge Distillation into Dense Architectures

We distill reasoning tokens from DeepSeek-R1 into smaller dense Qwen and Llama architectures (1.5B, 7B, 14B, 32B). Distillation yields superior results compared to training small models with RL from scratch.

4. Benchmark Performance & Mathematical Breakthroughs

DeepSeek-R1 records 79.8% pass@1 on AIME 2024 (matching OpenAI-o1-1217), 97.3% on MATH-500, and 2029 Elo on Codeforces (top 96.3th percentile). The distilled 32B model attains 72.6% on AIME 2024, surpassing all open-source models of comparable scale.

5. Limitations and Reasoning Vulnerabilities

DeepSeek-R1 possesses clear limitations: First, inference latency is high due to long thinking traces (often several thousand reasoning tokens per query). Second, while strong in verifiable STEM domains with rule-based ground-truth rewards, applying GRPO to creative writing or subjective policy tasks remains challenging due to lack of deterministic reward verifiers. Third, overthinking on simple factual queries can lead to unnecessary computational overhead.

6. Future Research Trajectories

Key frontiers include dynamic thinking token budget allocation, extending pure RL reasoning to multi-modal visual proofs, and eliminating reasoning token verbosity through length penalties.